Notes: Duflo & Saez (QJE, 2003)

The Role of Information and Social Interactions in Retirement Plan Decisions: Evidence from a Randomized Experiment

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1 Big picture

- **Question.** Do information and peer interactions affect employees' decisions to attend a benefits fair and enroll in a Tax Deferred Account retirement plan?
- **Setting.** A large university holds an annual benefits fair during open enrollment. The fair provides information about health benefits, retirement benefits, mutual fund vendors, and the university's TDA plan.
- **Experiment.** The authors run a randomized encouragement design. Some departments are randomly treated. Within treated departments, half of initially non-enrolled staff receive a letter offering a $20 reward for attending the fair.
- **Two treatments.** An employee can receive the individual letter treatment $L_{ij} = 1$, and an employee can belong to a treated department $D_j = 1$ where some colleagues receive the letter.
- **Main first-stage result.** The incentive letter dramatically raises fair attendance. Attendance is 28.0% among letter recipients, 15.1% among non-recipients in treated departments, and 4.9% in untreated departments.
- **Main enrollment result.** Treated departments have higher TDA enrollment after the fair. The reduced-form department effect is 0.93 percentage points after 4.5 months and 1.25 percentage points after 11 months.
- **Surprise.** Directly treated employees are not more likely to enroll than untreated employees in the same treated departments. The indirect department effect is almost as large as the direct letter effect, and sometimes larger.
- **Interpretation problem.** The result can be explained by three mechanisms: social-network effects, differential treatment effects across compliers, or motivational effects from paying people to attend.
- **Policy message.** Small interventions can change retirement-saving decisions through social channels, but the absolute enrollment effect is small compared with default enrollment policies.
- **Broader contribution.** The paper shows both the promise and the difficulty of randomized experiments in social settings: randomization identifies reduced-form effects cleanly, but separating channels is hard when people interact.

2 Data and measurement

2.1 Institutional setting

- The university offers staff employees a TDA plan, similar in spirit to a 401(k), in which employees choose whether to contribute.
- The benefits fair is held once per year, during open enrollment, at two locations close to the university's two main campuses.
- Employees receive a standard university benefits packet and invitation to the fair. The experiment adds a separate researcher/university letter to a randomly selected group.
- The fair includes the Benefits Office, health-plan providers, mutual-fund vendors, written materials, and a computer program to analyze the employee's situation.

Interpretation: The fair is an information intervention, but the experiment studies more than information. The letter changes attention, attendance, and the behavior of coworkers in the same department.

2.2 Sample construction

- The sample starts from all staff employees, not faculty, who are under age 65 and eligible for the TDA.
- About 9,700 employees meet the basic eligibility criteria.
- Around 3,500 were already enrolled in the TDA before the experiment.
- The main analysis focuses on about 6,200 employees not enrolled before the fair, because few employees stop contributing once enrolled.
- Departments are the social unit. The university has 330 departments, and the average department has about 30 staff employees.

Interpretation: The outcome of interest is new enrollment. Existing participants are less informative because the relevant decision is whether to start contributing.

2.3 Randomization design

The authors use a two-stage design.

1. **Department randomization.** Departments are grouped into triplets matched by size and pre-fair TDA participation. In each triplet, two of three departments are randomly assigned to treatment status $D_j = 1$.
2. **Individual randomization inside treated departments.** Among initially non-enrolled employees in treated departments, half are randomly selected to receive the letter and $20 attendance incentive, so $L_{ij} = 1$.

The three observed groups are:

- group 11: treated department and individual letter, $D_j = 1, L_{ij} = 1$;
- group 10: treated department and no individual letter, $D_j = 1, L_{ij} = 0$;
- group 00: untreated department and no individual letter, $D_j = 0, L_{ij} = 0$.

Interpretation: The design creates clean reduced-form comparisons for both individual encouragement and department-level spillovers.

2.4 Outcome variables

Let f_{ij} be fair attendance and y_{ij} be post-fair TDA enrollment.

- **Fair attendance:** whether employee i in department j attended the benefits fair.
- **TDA enrollment after 4.5 months:** administrative record of whether the employee started contributing after the fair.
- **TDA enrollment after 11 months:** same outcome at the longer follow-up horizon.
- **Baseline controls:** gender, years of service, age, salary, and department-triplet fixed effects.

Interpretation: Administrative enrollment data are important because they avoid survey response bias. Attendance data are less complete for employees who did not receive the letter, which is why some attendance measures require adjustment.

3 Models and empirical specifications

3.1 Average department treatment effects

The first reduced-form specification compares treated and untreated departments:

$$f_{ij} = \alpha_1 + \beta_1 D_j + \varepsilon_{ij}, \tag{1}$$

$$y_{ij} = \alpha_2 + \beta_2 D_j + \eta_{ij}. \tag{2}$$

Interpretation: β_1 measures the effect of being in a department where some employees were encouraged to attend the fair. β_2 measures the corresponding effect on TDA enrollment. These are clean department-level reduced-form effects.

3.2 Separating individual and department treatment

The second reduced-form specification includes both the individual letter and the department treatment:

$$f_{ij} = \alpha_1 + \mu_1 L_{ij} + \delta_1 D_j + \varepsilon_{ij}, \tag{3}$$

$$y_{ij} = \alpha_2 + \mu_2 L_{ij} + \delta_2 D_j + \eta_{ij}. \tag{4}$$

Interpretation: μ captures the extra effect of personally receiving the letter, conditional on being in a treated department. δ captures the indirect or spillover effect of being in a treated department without personally receiving the letter.

3.3 Treatment-effect decomposition

A more structural interpretation writes enrollment as

$$y_{ij} = \alpha + \gamma_i f_{ij} + \Gamma D_j + u_{ij}. \tag{5}$$

- γ_i is the individual-level effect of attending the fair.
- Γ is the department-level social effect of being around coworkers exposed to the letter.
- D_j can capture information diffusion, peer pressure, or coworker reminders.

The authors also allow the letter to change the effect of fair attendance:

$$\gamma_i = \gamma_i^S - \nu L_{ij}, \tag{6}$$

where ν is a possible motivational reward effect.

Interpretation: If being paid to attend makes people treat the fair as a chore, the attendance induced by the \$20 reward may have a lower treatment effect than attendance induced by colleagues.

3.4 Potential attendance types

Let $f_{ij}(11)$, $f_{ij}(10)$, and $f_{ij}(00)$ be potential attendance decisions under the three experimental states. The monotonicity assumption is

$$f_{ij}(11) \geq f_{ij}(10) \geq f_{ij}(00). \tag{7}$$

This partitions employees into:

- **Never takers:** do not attend in any state.

- **Financial reward compliers:** attend only when personally offered the $20 reward.
- **Social interaction compliers:** attend when their department is treated, even without their own letter.
- **Always takers:** attend regardless of treatment.

Interpretation: The paper's central identification difficulty is that the two instruments, L_{ij} and D_j, cannot separately identify all treatment-effect heterogeneity, social effects, and motivational reward effects.

3.5 Alternative IV interpretations

If there is no motivational reward effect and the fair treatment effect is constant, the model becomes

$$y_{ij} = \alpha + \gamma f_{ij} + \Gamma D_j + u_{ij}, \tag{8}$$

with L_{ij} and D_j as instruments.

If instead there are no social effects, the model becomes

$$y_{ij} = \alpha + \gamma_i f_{ij} + u_{ij}, \tag{9}$$

and different instruments identify treatment effects for different complier groups.

Interpretation: The IV results are not a single definitive causal channel. They show how the interpretation changes under different assumptions.

4 Identification and empirical design

4.1 What randomization identifies cleanly

- Random assignment of departments identifies the effect of being in a treated department.
- Random assignment of letters within treated departments identifies the additional effect of personally receiving the reward letter.
- Matching departments into triplets by size and baseline participation improves precision and balance.
- Administrative data allow follow-up for enrollment without relying on self-reports.

Interpretation: The reduced-form effects in Table II are credible experimental estimates.

4.2 Why peer effects are hard

- A treated coworker can directly remind someone about the fair.
- Coworkers can attend in groups.
- Information obtained at the fair can diffuse through conversations.
- Coworker enrollment itself can affect beliefs, norms, or salience.
- The experiment observes the social unit at the department level but not the exact network graph.

Interpretation: The design identifies spillovers, but it does not distinguish information transmission, social pressure, and attention effects.

4.3 Why the direct letter effect is ambiguous

- The letter includes a financial incentive, so it induces attendance among people who may have low initial interest in the fair.
- Paying people may reduce intrinsic motivation or make the event feel less personally relevant.
- Therefore, personally receiving the letter may raise attendance without raising enrollment much.

Interpretation: The absence of a strong individual letter effect on TDA enrollment does not necessarily mean fair attendance has no value. It may mean the marginal attendees induced by cash are different or less responsive.

4.4 Main limitation

- The experiment supplies two instruments, L_{ij} and D_j, but the behavioral model contains more than two mechanisms.
- As a result, the authors present alternative interpretations and bounds rather than claiming that one exact channel is proven.
- Attendance for some non-letter employees is measured at an aggregate level, not with complete individual identities.

Interpretation: The paper is strongest as evidence that social context matters; it is more cautious about the precise reason social context matters.

5 Tables and figures

This paper has no original figures. The original tables below are directly cropped from the uploaded paper and grouped compactly to speed up reading.

5.1 Table I: Descriptive statistics by groups

Read:

- The pre-fair TDA participation rate is nearly identical across groups, around 1%, confirming balance before the fair for the main sample.
- Demographics are also well balanced: sex, years of service, annual salary, and age are very similar across treatment groups.
- Fair attendance among initially non-enrolled employees is 28.0% for group 11, 15.1% for group 10, and 4.9% for group 00.
- TDA enrollment after 4.5 months is 4.5% in group 11, 5.3% in group 10, and 4.0% in group 00.
- TDA enrollment after 11 months is 8.9% in group 11, 8.8% in group 10, and 7.5% in group 00.

Economic message: The experiment strongly raises fair attendance, including among untreated coworkers. The enrollment effect comes mainly from department exposure, not from individual letter status.

TABLE I
Descriptive Statistics, by Groups

	Treated departments			
	All (group $D = 1$)	Treated (group $D = 1$, $L = 1$)	Untreated (group $D = 1$, $L = 0$)	Untreated departments (group $D = 0$)
	(1)	(2)	(3)	(4)
PANEL A: BACKGROUND CHARACTERISTICS				
TDA participation before	0.010	0.009	0.011	0.012
the fair (Sept. 2000)	(.0015)	(.0021)	(.0022)	(.0024)
Observations	4168	2039	2129	2043
Sex (fraction male)	0.398	0.400	0.396	0.418
	(.0076)	(.0109)	(.0107)	(.011)
Years of service	5.898	5.864	5.930	6.008
	(.114)	(.161)	(.16)	(.157)
Annual salary	38,547	38,807	38,297	38,213
	(304)	(438)	(422)	(416)
Age	38.3	38.4	38.2	38.7
	(.17)	(.24)	(.24)	(.24)
Observations	4126	2020	2106	2018
PANEL B: FAIR ATTENDANCE (REGISTRATION DATA)				
Fair attendance rate among	0.214	0.280	0.151	0.049
non-TDA enrollees	(.0064)	(.01)	(.0078)	(.0048)
Observations	4126	2020	2106	2018
Fair attendance rate for all	0.192			0.063
staff employees	(.0132)			(.0103)
Observations	6687			3311
PANEL C: TDA PARTICIPATION (ADMINISTRATIVE DATA)				
TDA participation rate after	0.049	0.045	0.053	0.040
4.5 months	(.0035)	(.0049)	(.0051)	(.0045)
Observations	3726	1832	1894	1861
TDA participation rate after	0.088	0.089	0.088	0.075
11 months	(.005)	(.0071)	(.007)	(.0065)
Observations	3246	1608	1638	1633

a. Standard errors are in parentheses.
b. The first part of Panel B includes all individuals not enrolled in the TDA by September 2000. The second part includes all employees (enrolled or not in the TDA).
c. The average fair participation in the nontreated departments was obtained from the registration information collected at the fair. Since only 75 percent of the participants registered, the participation was adjusted by a proportionality factor.
d. Demographic information and TDA participation are all obtained from administrative data.

TABLE II
Reduced-Form Estimates (OLS)

	Dependent variable		
		TDA enrollment after	
	Fair attendance	4.5 months	11 months
	(1)	(2)	(3)
PANEL A: Average effect of department treatment			
Treated	0.166	0.0093	0.0125
Department dummy D	(.013)	(.0043)	(.0065)
Observations	6144	5587	4879
PANEL B: Effect of letter and department treatment			
Letter dummy L	0.129	−0.0066	0.0005
	(.0226)	(.0061)	(.0102)
Treated	0.102	0.0125	0.0123
Department dummy D	(.0139)	(.0054)	(.0086)
Observations	6144	5587	4879

a. Dependent variables are individual fair participation (column (1)), TDA enrollment 4.5 months and 11 months after the fair (columns (2) and (3)).
b. Independent variable in Panel A is the department treatment dummy D.
c. Independent variables in Panel B are the individual letter dummy L and the department treatment dummy D.
d. All regressions control for the triplet of the department, gender, year of service, age, and salary.
e. Standard errors (in parentheses) are corrected for clustering at the department level.

5.2 Tables II and III: Reduced form and IV interpretations

Read: Table II:

- Department treatment raises fair attendance by 16.6 percentage points.
- Department treatment raises TDA enrollment by 0.93 percentage points after 4.5 months and 1.25 percentage points after 11 months.
- Receiving the letter personally raises attendance by 12.9 percentage points, but has essentially no positive effect on TDA enrollment.
- Being in a treated department without necessarily receiving a letter raises attendance by 10.2 percentage points and raises TDA enrollment by about 1.2 percentage points.

Read: Table III:

- Under constant treatment effects, IV estimates imply little direct fair-attendance effect and a positive department social effect.
- Under no-social-effects assumptions, the estimated effect for financial-reward compliers is near zero, while the effect for social-interaction compliers is 11.7 percentage points after 4.5 months and 13.1 percentage points after 11 months.
- OLS among letter recipients gives a positive 4.9 percentage-point estimate after 11 months, but the authors interpret this as likely upward-biased by selection.

Economic message: The reduced form is clean: treated departments enroll more. The channel is less clean: cash-induced attendance is not the same as socially induced attendance.

5.3 Table IV: Heterogeneous effects

Read:

- Fair attendance among letter recipients is higher in smaller departments, among women, and among lower-tenure employees.
- The TDA enrollment effects are imprecise across subgroups.
- There is no strong evidence that the final 11-month effects differ sharply by department size, baseline participation, gender, salary, or years of service.

Economic message: The evidence points to a broad department-level effect rather than a narrow treatment effect concentrated in one observable subgroup.

TABLE III
IV Estimates of Fair Attendance and Department Effects on TDA Enrollment

	Assuming constant treatment effect	Assuming no social effects: Effect on financial incentive compliers	Assuming no social effects: Effect on social interaction compliers	OLS	Naïve IV
	(1)	(2)	(3)	(4)	(5)
PANEL A: Dependent variable: TDA participation after 4.5 months					
Fair attendance	−0.046	−0.050	0.117	0.016	−0.002
	(.0431)	(.0429)	(.0465)	(.0109)	(.0255)
Treated department	0.018				
	(.0092)				
Observations	5587	3726	3755	1832	5587
PANEL B: Dependent variable: TDA participation after 11 months					
Fair attendance	0.003	0.005	0.131	0.049	0.032
	(.0681)	(.0685)	(.0826)	(.018)	(.0397)
Treated department	0.012				
	(.0147)				
Observations	4879	3246	3271	1608	4879
Sample	Entire sample	Treated departments	No letter only	Letter only	Entire sample

a. Dependent variables are individual enrollment in the TDA 4.5 months and 11 months after the fair.
b. Independent variable are individual fair attendance and department treatment dummy D in column (1).
c. Independent variable is individual fair attendance in columns (2) to (5).
d. All regressions control for the triplet of the department, gender, year of service, age, and salary.
e. Standard errors (in parentheses) are corrected for clustering at the department level.

TABLE IV
Fair Attendance and Treatment Effect in Different Groups

	Fair attendance among letter recipients ($L = 1$)	Difference group $D = 1$ − group $D = 0$: TDA participation after 4.5 months	Difference group $D = 1$ − group $D = 0$: TDA participation after 11 months
	(1)	(2)	(3)
PANEL A: DEPARTMENT SIZE			
Below median (81)	0.328	0.008	0.007
	(.015)	(.007)	(.0104)
Observations	985	2797	2403
Above median (81)	0.235	0.007	0.012
	(.0132)	(.0047)	(.0087)
Observations	1035	2790	2476
PANEL B: DEPARTMENT AVERAGE PARTICIPATION IN THE TDA BEFORE THE EXPERIMENT			
Below median (34%)	0.259	0.009	0.018
	(.0134)	(.0064)	(.0098)
Observations	1062	2929	2523
Above median (34%)	0.304	0.010	0.008
	(.0149)	(.0063)	(.0089)
Observations	958	2658	2356
PANEL C: GENDER			
Women	0.320	0.011	0.012
	(.0134)	(.0072)	(.0117)
Observations	1213	3298	2843
Men	0.221	0.008	0.010
	(.0146)	(.007)	(.0085)
Observations	807	2289	2036
PANEL D: SALARY			
Below median ($34,021)	0.269	0.003	0.018
	(.0141)	(.0065)	(.0093)
Observations	983	2745	2291
Above median ($34,021)	0.291	0.015	0.010
	(.0141)	(.0063)	(.0104)
Observations	1037	2842	2588
PANEL E: YEARS OF SERVICE			
Below median (2.84 years)	0.312	0.005	0.015
	(.0145)	(.0071)	(.0115)
Observations	1027	2706	2196
Above median (2.84 years)	0.248	0.013	0.010
	(.0137)	(.0054)	(.0083)
Observations	993	2881	2683

a. The sample in column (1) is composed of individuals in group 11 only.
b. In columns (2) and (3), dependent variables are TDA enrollment 4.5 months and 11 months after the fair. Independent variable is department treatment dummy D.
c. Regressions in columns (2) and (3) control for the triplet of the department, gender, year of service, age, and salary.
d. Standard errors (reported in parentheses below the coefficient) are corrected for clustering at the department level.

6 Short summary

- The paper studies a randomized intervention designed to increase attendance at a university benefits fair and TDA enrollment.
- The experiment has two layers: department-level treatment and individual-level letter treatment.
- Group 11 receives the letter and works in a treated department; group 10 works in a treated department but receives no letter; group 00 is in an untreated department.
- The $20 letter strongly increases fair attendance.
- Coworker exposure also strongly increases fair attendance: non-letter employees in treated departments attend much more than employees in untreated departments.
- Treated departments have higher TDA enrollment after both 4.5 and 11 months.
- Personal letter receipt does not produce a larger enrollment effect than coworker exposure.
- The authors propose three explanations: social networks, differential treatment effects, and motivational reward effects.
- The data clearly show spillovers, but do not fully separate the mechanism behind those spillovers.
- The intervention is cheap relative to the additional TDA contributions it induces, but its absolute impact is small compared with automatic enrollment or default-based policies.

7 Conclusion

7.1 Core conclusion

Duflo and Saez show that small randomized information and incentive interventions can affect retirement-plan decisions, especially through social channels inside the workplace. The key empirical fact is that department-level exposure matters for both fair attendance and TDA enrollment, while personal receipt of the monetary reward letter has little additional effect on enrollment.

7.2 Economic intuition

Retirement saving is complicated, and many employees may not actively seek information. A small reward can make some people attend a fair, but coworker exposure can also increase attention, discussion, and salience. Once some coworkers attend or talk about the fair, others may update beliefs, feel encouraged, or simply remember to act. The relevant intervention therefore operates through a social environment, not only through private information delivery.

7.3 Takeaway

The paper is a classic example of using randomized encouragement to study social interactions. Its lesson is not only that financial education can matter, but that the pathway from information to behavior is socially mediated and empirically hard to decompose. For policy, the results support active outreach and peer-based diffusion, while also showing that default rules may be far more powerful than information sessions alone.